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Paper Code: TBT-507

END SEMESTER EXAMINATION, DECEMBER 2025

B. Tech Biotechnology, Semester V

Biosimilar Technology

Time: 3 Hr

MM: 100

Note:

- (i) Answer **all the questions** by choosing *any two of the sub questions*.
(ii) Each question carries 10 marks

Q1	(10×2 = 20 marks)	
(a)	Elaborate the ways and application of chemical genomics in biosimilar development.	CO1
(b)	Explain the regulatory guidelines and its need for biosimilar development.	
(c)	Describe stepwise the prerequisites of biosimilar and conditions for its approval.	
Q2	(10×2 = 20 marks)	
(a)	Write a note on: cell biologics and explain at least four examples of it	CO2
(b)	What is gene therapy? Give examples of FDA approved gene therapies and their indications.	
(c)	"Biosimilars reduces the cost of biologics many fold" Explain using any example.	
Q3	(10×2 = 20 marks)	
(a)	Describe the techniques used for adsorption of protein and peptides on different surfaces.	CO3
(b)	What are the methods of protein sequencing? Explain them.	
(c)	What are the factors that impact protein stability and how the stability can be enhanced.	
Q4	(10×2 = 20 marks)	
(a)	Discuss the experimental design and statistical issues involved in bioequivalence studies.	CO4
(b)	What are the requirements for a biosimilar to be designated inter-changeable by the FDA?	
(c)	Elucidate the role and basis of ORANGE BOOK.	
Q5	(10×2 = 20 marks)	
(a)	Discuss the importance and application of monoclonal antibodies.	CO5
(b)	Highlight a case study involving the use of streptokinase in acute myocardial infarction.	
(c)	Discuss the insulin related biosimilars prepared by biotech companies.	

